

MATH 152 — PART 2

Sequences
Detailed Notes with Tips and Examples

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2.1 Definition of a Sequence

What is a Sequence?

A **sequence** is a set of numbers arranged in a **definite order** according to a **definite rule**.

Key Characteristics:

- Order matters! $(1, 2, 3) \neq (3, 2, 1)$
- Each number is called a **term**
- Terms are denoted by: $U_1, U_2, U_3, \dots, U_n$
- The sequence itself is denoted by $\{U_n\}$

Types of Sequences

Type	Definition	Example
Finite Sequence	Has a last term	$\{1, 2, 3, \dots, 20\}$
Infinite Sequence	No last term	$\{1, 1/2, 1/3, 1/4, \dots\}$

 **TIP:** The n th term U_n gives the value of any term in the sequence.

2.2 Types of Sequences

2.2.1 Arithmetic Sequence (AP)

Definition

A sequence is **arithmetic** if the difference between consecutive terms is **constant**.

$$U_{n+1} - U_n = d, \text{ for all } n$$

d is called the **common difference**.

Examples

Sequence	a (first term)	d (common difference)
2, 5, 8, 11, 14, ...	2	3
10, 7, 4, 1, -2, ...	10	-3
$1/2, 1, 3/2, 2, \dots$	$1/2$	$1/2$

General Term (nth Term)

$$U_n = a + (n - 1)d$$

Example: For 2, 5, 8, 11, ...

$$U_5 = 2 + (5-1)(3) = 2 + 12 = 14 \checkmark$$

Sum of First n Terms

$$S_n = n/2 \cdot [2a + (n - 1)d]$$

Alternative Form:

$$S_n = n/2 \cdot (a + U_n)$$

Example: Sum of first 10 terms of 2, 5, 8, 11, ...

$$\begin{aligned} S_{10} &= 10/2 \cdot [2(2) + (10-1)(3)] \\ &= 5 \cdot [4 + 27] \\ &= 5 \times 31 \\ &= 155 \end{aligned}$$

2.2.2 Geometric Sequence (GP)

Definition

A sequence is **geometric** if the ratio between consecutive terms is **constant**.

$$U_{n+1} / U_n = r, \text{ for all } n$$

r is called the **common ratio**.

Examples

Sequence	a (first term)	r (common ratio)
1, 2, 4, 8, 16, ...	1	2
1, 1/2, 1/4, 1/8, ...	1	1/2
27, 9, 3, 1, 1/3, ...	27	1/3
2, -4, 8, -16, ...	2	-2

General Term (nth Term)

$$U_n = a \times r^{(n-1)}$$

Example: For 1, 2, 4, 8, ...

$$U_6 = 1 \times 2^5 = 32 \checkmark$$

Sum of First n Terms

$$\begin{aligned} S_n &= a(1 - r^n) / (1 - r) && \text{for } |r| < 1 \\ S_n &= a(r^n - 1) / (r - 1) && \text{for } |r| > 1 \end{aligned}$$

Example: Sum of first 6 terms of 1, 2, 4, 8, ...

$$S_6 = 1(2^6 - 1) / (2 - 1) = (64 - 1) / 1 = 63$$

Check: 1 + 2 + 4 + 8 + 16 + 32 = 63 ✓

Sum to Infinity

$$S_\infty = a / (1 - r) \quad \text{for } |r| < 1$$

Example: Sum to infinity of 1, 1/2, 1/4, 1/8, ...

$$S_\infty = 1 / (1 - 1/2) = 1 / (1/2) = 2$$

Visual Check: 1 + 0.5 + 0.25 + 0.125 + ... = 2 ✓

Important Note on r

Value of r	Behavior
r < 1	Converges (terms get smaller)
r > 1	Diverges (terms get larger)
r = 1	Constant sequence
r = -1	Alternates between a and -a
r = 1	Does not converge (except r=1 constant)

💡TIP: The sum to infinity formula only works when |r| < 1!

Quick Comparison: AP vs GP

Feature	Arithmetic (AP)	Geometric (GP)
Operation	Addition	Multiplication
Key value	d (common difference)	r (common ratio)
General term	$U_n = a + (n-1)d$	$U_n = a \times r^{(n-1)}$
Sum formula	$S_n = n/2[2a+(n-1)d]$	$S_n = a(1-r^n)/(1-r)$
Sum to infinity	Not applicable	$S_\infty = a/(1-r)$ if $ r <1$

2.3 Limit of a Sequence

Intuitive Definition

An infinite sequence $\{U_n\}$ has a **limit l** as n approaches infinity if U_n can be made **as close as we please** to l by choosing n **sufficiently large**.

$$\lim_{n \rightarrow \infty} U_n = 1$$

Formal (ϵ - δ) Definition

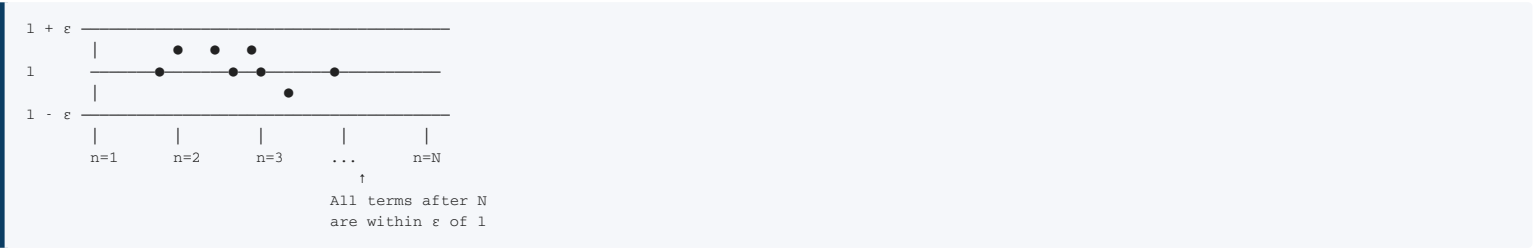
$$\lim_{n \rightarrow \infty} U_n = 1$$

if for **every** $\epsilon > 0$ (however small), there exists a positive integer **N** (depending on ϵ) such that:

$$|U_n - 1| < \epsilon, \text{ for all } n > N$$

🔑**KEY INSIGHT:** No matter how small you make ϵ , you can always find a term far enough in the sequence that gets closer than ϵ to l .

Visual Understanding



Examples of Limits

Example 1: $\{1/n\}$

$$\lim_{n \rightarrow \infty} 1/n = 0$$

Proof:

For any $\epsilon > 0$, we need $|1/n - 0| < \epsilon$
 $\Rightarrow 1/n < \epsilon$
 $\Rightarrow n > 1/\epsilon$

Choose $N = \text{any integer} > 1/\epsilon$.
Then for all $n > N$, $|1/n - 0| < \epsilon$.

Therefore, $\lim_{n \rightarrow \infty} 1/n = 0$ ✓

Example 2: $\{(n+1)/n\}$

$$\lim_{n \rightarrow \infty} (n+1)/n = \lim_{n \rightarrow \infty} (1 + 1/n) = 1 + 0 = 1$$

Terms:

$n=1$: $2/1 = 2$
 $n=2$: $3/2 = 1.5$
 $n=3$: $4/3 = 1.333$
 $n=4$: $5/4 = 1.25$
 $n=10$: $11/10 = 1.1$
 $n=100$: $101/100 = 1.01$

As n grows, terms approach 1. ✓

Example 3: $\{(n^2 + 1)/n\}$

$$\lim_{n \rightarrow \infty} (n^2 + 1)/n = \lim_{n \rightarrow \infty} (n + 1/n) = \infty$$

Terms:

$n=1$: 2
 $n=2$: $5/2 = 2.5$
 $n=3$: $10/3 = 3.33$
 $n=10$: $101/10 = 10.1$

The terms grow without bound → **DIVERGENT**

Example 4: $\{(3/4)^n\}$

$$\lim_{n \rightarrow \infty} (3/4)^n = 0$$

Terms:

n=1:	0.75
n=2:	0.5625
n=3:	0.421875
n=10:	0.0563
n=20:	0.00317

Terms approach 0 → **CONVERGENT**

Example 5: $\{(-1)^n\}$

$$\lim_{n \rightarrow \infty} (-1)^n \text{ does not exist}$$

Terms:

n=1:	-1
n=2:	1
n=3:	-1
n=4:	1

Alternates between -1 and 1 → **DIVERGENT**

Terminology

Type	Definition
Convergent	Has a limit (finite)
Divergent	Does NOT have a limit
Diverges to ∞	Increases without bound
Diverges to $-\infty$	Decreases without bound
Oscillates	Neither approaches ∞ nor $-\infty$

💡**TIP:** A sequence can be divergent without going to infinity (e.g., $(-1)^n$ oscillates).

2.4 Theorems on Sequences

Theorem 1: Uniqueness of Limit

If a sequence $\{U_n\}$ has a limit, then the limit is **unique**.

PROOF

Assume $\{U_n\}$ has two limits l_1 and l_2 .

For any $\varepsilon > 0$:

- $\exists N_1$ such that $|U_n - l_1| < \varepsilon$ for $n > N_1$
- $\exists N_2$ such that $|U_n - l_2| < \varepsilon$ for $n > N_2$

Choose $n > \max(N_1, N_2)$:

$$\begin{aligned} |l_1 - l_2| &= |l_1 - U_n + U_n - l_2| \\ &\leq |l_1 - U_n| + |U_n - l_2| \\ &< \varepsilon + \varepsilon = 2\varepsilon \end{aligned}$$

Since ε can be arbitrarily small, $|l_1 - l_2| = 0$.
Therefore, $l_1 = l_2$.

This proves the limit is unique!

Theorem 2: Algebra of Limits

If $\lim_{n \rightarrow \infty} U_n = A$ and $\lim_{n \rightarrow \infty} V_n = B$, then:

(a) Sum Rule

$$\lim_{n \rightarrow \infty} (U_n + V_n) = A + B$$

Example: $\lim_{n \rightarrow \infty} 1/n + \lim_{n \rightarrow \infty} 1/n^2 = 0 + 0 = 0$

(b) Difference Rule

$$\lim_{n \rightarrow \infty} (U_n - V_n) = A - B$$

(c) Product Rule

$$\lim_{n \rightarrow \infty} (U_n \times V_n) = A \times B$$

(d) Quotient Rule

$$\lim_{n \rightarrow \infty} (U_n / V_n) = A / B, \text{ provided } B \neq 0$$

Special Cases:

- If $B = 0$ and $A \neq 0 \rightarrow$ limit does not exist
- If $B = 0$ and $A = 0 \rightarrow$ limit may or may not exist (indeterminate)

(e) Power Rule

$$\lim_{n \rightarrow \infty} U_n^p = A^p \text{ (where } p \text{ is real and } A^p \text{ exists)}$$

(f) Exponential Rule

$$\lim_{n \rightarrow \infty} p^{U_n} = p^A \text{ (where } p \text{ is real and } p^A \text{ exists)}$$

Applications of Limit Theorems

EXAMPLE 1

$$\lim_{n \rightarrow \infty} (4 - 2n - 3n^2) / (4n^2 + 5n)$$

Step 1: Divide numerator and denominator by n^2

$$= \lim_{n \rightarrow \infty} (4/n^2 - 2/n - 3) / (4 + 5/n)$$

Step 2: Apply limit to each term

$$= (0 - 0 - 3) / (4 + 0) = -3/4$$

EXAMPLE 2

$$\lim_{n \rightarrow \infty} \sqrt{n^2(n-1)/(1 + \frac{1}{2}n^3)}$$

Step 1: Simplify inside the root
$$= \lim_{n \rightarrow \infty} \sqrt{(n^3 - n^2)/(1 + \frac{1}{2}n^3)}$$

Step 2: Divide numerator and denominator by n^3
$$= \lim_{n \rightarrow \infty} \sqrt{((1 - 1/n)/(1/n^3 + \frac{1}{2}))}$$

Step 3: Apply limit
$$= \sqrt{(1 - 0)/(0 + \frac{1}{2})} = \sqrt{1/\frac{1}{2}} = \sqrt{2}$$

EXAMPLE 3

$$\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$$

Step 1: Rationalize
$$= \lim_{n \rightarrow \infty} [(\sqrt{n^2+n} - n)(\sqrt{n^2+n} + n)/(\sqrt{n^2+n} + n)]$$
$$= \lim_{n \rightarrow \infty} [(n^2+n - n^2)/(\sqrt{n^2+n} + n)]$$
$$= \lim_{n \rightarrow \infty} [n/(\sqrt{n^2+n} + n)]$$

Step 2: Divide by n
$$= \lim_{n \rightarrow \infty} [1/(\sqrt{1 + 1/n} + 1)]$$
$$= 1/(\sqrt{1} + 1) = 1/2$$

EXAMPLE 4

$$\lim_{n \rightarrow \infty} [n^2/2^{(n+1)}]$$

Let $U_n = n^2/2^{(n+1)}$

We can use L'Hospital's rule for sequences
(or note that exponential dominates polynomial)

$$= 0 \text{ (since } 2^n \text{ grows much faster than } n^2 \text{)}$$

Indeterminate Forms

Form	Example	What to Do
0/0	$\lim_{n \rightarrow \infty} (n-1)/(n^2-1)$	Factor and cancel
∞/∞	$\lim_{n \rightarrow \infty} (n^2+1)/(3n^2+n)$	Divide by highest power
$0 \times \infty$	$\lim_{n \rightarrow \infty} n \cdot (1/n^2)$	Rewrite as fraction
$\infty - \infty$	$\lim_{n \rightarrow \infty} (n+1 - n)$	Combine terms
1^∞	$\lim_{n \rightarrow \infty} (1+1/n)^n$	Use e formula

🔑KEY INSIGHT: When you see any of these, you CANNOT just substitute $n = \infty$. You must manipulate the expression first!

2.5 Bounded and Monotonic Sequences

Bounded Sequences

Type	Definition
Bounded above	$\exists M$ such that $U_n \leq M, \forall n$
Bounded below	$\exists m$ such that $U_n \geq m, \forall n$
Bounded	$\exists m, M$ such that $m \leq U_n \leq M, \forall n$

Least Upper Bound (l.u.b.) and Greatest Lower Bound (g.l.b.)

- **l.u.b.** = smallest upper bound
- **g.l.b.** = largest lower bound

Example: $U_n = 2 + 1/n$

Terms: 3, 2.5, 2.333, 2.25, 2.2, ...

Upper bound: 3, 4, 5, ...
l.u.b. = 3 (the first term)

Lower bound: 2, 1, 0, ...
g.l.b. = 2 (the limit as $n \rightarrow \infty$)

So the sequence is bounded.

Monotonic Sequences

Type	Definition
Monotonic increasing	$U_n \leq U_{n+1}, \forall n$ (non-decreasing)
Strictly increasing	$U_n < U_{n+1}, \forall n$
Monotonic decreasing	$U_n \geq U_{n+1}, \forall n$ (non-increasing)
Strictly decreasing	$U_n > U_{n+1}, \forall n$

EXAMPLE 1 — Check if $\{(n+1)/n\}$ is monotonic

$$U_n = (n+1)/n = 1 + 1/n$$

Check $U_{n+1} - U_n$:

$$\begin{aligned} U_{n+1} - U_n &= (1 + 1/(n+1)) - (1 + 1/n) \\ &= 1/(n+1) - 1/n \\ &= (n - (n+1))/(n(n+1)) \\ &= -1/[n(n+1)] < 0 \end{aligned}$$

Since $U_{n+1} - U_n < 0$, $U_{n+1} < U_n$.
Therefore, the sequence is strictly decreasing.

EXAMPLE 2 — Check if $\{n/(n+1)\}$ is monotonic

$$U_n = n/(n+1)$$

Check $U_{n+1} - U_n$:

$$\begin{aligned} U_{n+1} - U_n &= (n+1)/(n+2) - n/(n+1) \\ &= [(n+1)^2 - n(n+2)]/[(n+1)(n+2)] \\ &= [n^2 + 2n + 1 - n^2 - 2n]/[(n+1)(n+2)] \\ &= 1/[(n+1)(n+2)] > 0 \end{aligned}$$

Since $U_{n+1} - U_n > 0$, $U_{n+1} > U_n$.
Therefore, the sequence is strictly increasing.

Key Theorem: Bounded Monotonic Sequences

➡ **KEY INSIGHT:** Every bounded monotonic sequence has a limit.

Bounded + Monotonic $\Rightarrow$ Convergent

Why This is Important:

- If you can show a sequence is both bounded and monotonic, you know it converges without finding the actual limit!
- This is a powerful existence theorem.

Example: Proving Convergence Using the Theorem

PROBLEM — Show that {n/(n+1)} converges

Step 1: Show it's bounded

0 < n/(n+1) < 1 for all n
So bounded (above by 1, below by 0)

Step 2: Show it's monotonic (increasing)

U_{n+1} - U_n = 1/[(n+1)(n+2)] > 0
So strictly increasing

Step 3: Apply theorem

Bounded + Increasing ⇒ Convergent

Step 4: Find the limit (optional)

lim_(n→∞) n/(n+1) = lim_(n→∞) 1/(1 + 1/n) = 1

Detailed Example: Proving Limit of {n/(n+1)}

PROVE THAT lim_(n→∞) n/(n+1) = 1

Step 1: We want to show for every ε > 0, ∃ N such that:

|n/(n+1) - 1| < ε, ∀ n > N

Step 2: Simplify the expression:

|n/(n+1) - 1| = |n - (n+1)|/(n+1) = |-1|/(n+1) = 1/(n+1)

Step 3: We need 1/(n+1) < ε

1/(n+1) < ε
⇒ n+1 > 1/ε
⇒ n > 1/ε - 1

Step 4: Choose N as any integer > 1/ε - 1.

Step 5: For all n > N, we have:

|n/(n+1) - 1| = 1/(n+1) < ε

Therefore, lim_(n→∞) n/(n+1) = 1 ✓

More Limit Proofs

PROVE lim_(n→∞) (4n-3)/(2n+3) = 2

We need to show: For any ε > 0, ∃ N such that:

|(4n-3)/(2n+3) - 2| < ε, ∀ n > N

Simplify:

|(4n-3)/(2n+3) - 2| = |(4n-3 - 2(2n+3))/(2n+3)|
= |(4n-3 - 4n - 6)/(2n+3)|
= |-9/(2n+3)|
= 9/(2n+3)

We need 9/(2n+3) < ε:

9/(2n+3) < ε
⇒ 2n+3 > 9/ε
⇒ 2n > 9/ε - 3
⇒ n > (9/ε - 3)/2

Choose N > (9/ε - 3)/2.

Then for all n > N: |(4n-3)/(2n+3) - 2| < ε.

Therefore, lim_(n→∞) (4n-3)/(2n+3) = 2 ✓

PROVE $\lim_{n \rightarrow \infty} 2^{(1/n)} = 1$

We need to show: For any $\varepsilon > 0$, $\exists N$ such that:

$$|2^{(1/n)} - 1| < \varepsilon, \quad \forall n > N$$

Since $n \geq 1$, $2^{(1/n)} \geq 1$ (so no absolute value needed).

$$\begin{aligned} \text{We need: } 2^{(1/n)} - 1 &< \varepsilon \\ \Rightarrow 2^{(1/n)} &< 1 + \varepsilon \\ \Rightarrow (1/n) \log_2(2) &< \log_2(1 + \varepsilon) \\ \Rightarrow 1/n &< \log_2(1 + \varepsilon) \\ \Rightarrow n &> 1/\log_2(1 + \varepsilon) \end{aligned}$$

Choose $N > 1/\log_2(1 + \varepsilon)$.

Then for all $n > N$: $|2^{(1/n)} - 1| < \varepsilon$.

Therefore, $\lim_{n \rightarrow \infty} 2^{(1/n)} = 1$ ✓

PROVE $\lim_{n \rightarrow \infty} (3/4)^n = 0$

We need to show: For any $\varepsilon > 0$, $\exists N$ such that:

$$|(3/4)^n - 0| < \varepsilon, \quad \forall n > N$$

We need: $(3/4)^n < \varepsilon$

Take natural log of both sides:

$$\begin{aligned} \ln[(3/4)^n] &< \ln(\varepsilon) \\ n \cdot \ln(3/4) &< \ln(\varepsilon) \end{aligned}$$

Since $\ln(3/4) < 0$, dividing by it flips the inequality:

$$n > \ln(\varepsilon) / \ln(3/4)$$

Choose $N > \ln(\varepsilon) / \ln(3/4)$.

Then for all $n > N$: $|(3/4)^n - 0| < \varepsilon$.

Therefore, $\lim_{n \rightarrow \infty} (3/4)^n = 0$ ✓

💡**TIP:** When dealing with exponentials, taking logs is often the easiest approach!

2.6 Evaluating Limits of Sequences

General Strategy

Step	Action
1	Identify highest power of n in numerator/denominator
2	Divide numerator and denominator by highest power
3	Apply limit: terms with n in denominator go to 0
4	For expressions with roots, rationalize if needed
5	For 1^∞ forms, use e formula

Examples with Solutions

EXAMPLE 1 — Polynomial/Polynomial

$$\lim_{n \rightarrow \infty} (5n^2 + 3n + 3) / (7n^2 - n + 2)$$

Divide by n^2 :

$$= \lim_{n \rightarrow \infty} (5 + 3/n + 3/n^2) / (7 - 1/n + 2/n^2)$$
$$= (5 + 0 + 0) / (7 - 0 + 0) = 5/7$$

🔗**TIP:** For rational functions with equal degrees, the limit is the ratio of leading coefficients.

EXAMPLE 2 — Lower Degree Numerator

$$\lim_{n \rightarrow \infty} (3n^2 + 4n - 5) / (2n^3 - 6n^2 + n + 1)$$

Divide by n^3 :

$$= \lim_{n \rightarrow \infty} (3/n + 4/n^2 - 5/n^3) / (2 - 6/n + 1/n^2 + 1/n^3)$$
$$= (0 + 0 - 0) / (2 - 0 + 0 + 0) = 0$$

🔗**TIP:** If numerator degree < denominator degree, limit = 0.

EXAMPLE 3 — Higher Degree Numerator

$$\lim_{n \rightarrow \infty} (n^3) / (n^3 + 3)$$

Divide by n^3 :

$$= \lim_{n \rightarrow \infty} 1 / (1 + 3/n^3)$$
$$= 1 / (1 + 0) = 1$$

🔗**TIP:** If numerator degree = denominator degree, limit is ratio of leading coefficients.

EXAMPLE 4 — With Roots

$$\lim_{n \rightarrow \infty} \sqrt{3n^2 - 4n + 5} / (n + 1)$$

Divide numerator and denominator by n:

$$= \lim_{n \rightarrow \infty} \sqrt{3 - 4/n + 5/n^2} / (1 + 1/n)$$
$$= \sqrt{3 - 0 + 0} / (1 + 0) = \sqrt{3}$$

EXAMPLE 5 — Exponential

$$\lim_{n \rightarrow \infty} 5^{\sqrt{1 + 1/n^2}}$$

As $n \rightarrow \infty$, $1/n^2 \rightarrow 0$
So $\sqrt{1 + 1/n^2} \rightarrow \sqrt{1} = 1$
Therefore, $5^1 = 5$

EXAMPLE 6 — Summation to Infinity

$$\begin{aligned} &\lim_{n \rightarrow \infty} \left(\frac{1}{n^2} \right) \left(1 - \frac{1}{n^2} \right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n^2} - \lim_{n \rightarrow \infty} \frac{1}{n^4} \\ &= 0 - 0 = 0 \end{aligned}$$

Common Special Limits to Memorize

Limit	Value
$\lim_{n \rightarrow \infty} 1/n^k \ (k>0)$	0
$\lim_{n \rightarrow \infty} a^n \ (a <1)$	0
$\lim_{n \rightarrow \infty} n^b/a^n \ (a>1)$	0
$\lim_{n \rightarrow \infty} (1 + 1/n)^n$	e
$\lim_{n \rightarrow \infty} n^{1/n}$	1
$\lim_{n \rightarrow \infty} (1 + a/n)^n$	e ^a

2.7 Quick Reference: Convergence Tests

When Does a Sequence Converge?

Test	Condition	Result
Bounded Monotonic	Bounded + Monotonic	Converges
Geometric	r < 1	Converges to 0
Geometric	r = 1	Constant (converges)
Geometric	r ≥ 1	Diverges (except r=1)
Squeeze Theorem	a _n ≤ b _n ≤ c _n and a _n ,c _n → L	b _n → L

Practice Problems and Solutions

PROBLEM 1

Find the sum of the first 15 terms of 2, 5, 8, 11, ...

Solution:

```
a = 2, d = 3, n = 15

S15 = 15/2 [2(2) + (15-1)(3)]
      = 15/2 [4 + 42]
      = 15/2 × 46
      = 15 × 23
      = 345
```

PROBLEM 2

Find the sum to infinity of 1, 1/2, 1/4, 1/8, ...

Solution:

```
a = 1, r = 1/2
|r| = 1/2 < 1 ✓

S∞ = a/(1 - r) = 1/(1 - 1/2) = 1/(1/2) = 2
```

PROBLEM 3

Find the sum of the first 8 terms of 3, -6, 12, -24, ...

Solution:

```
a = 3, r = -2
|r| = 2 > 1

S8 = a(r^8 - 1)/(r - 1)
    = 3[(-2)^8 - 1]/(-2 - 1)
    = 3[256 - 1]/(-3)
    = 3 × 255/(-3)
    = -255
```

Check: 3 + (-6) + 12 + (-24) + 48 + (-96) + 192 + (-384) = -255 ✓

PROBLEM 4

Prove that the sequence {(n+1)/(3n-2)} converges to 1/3.

Solution:

```
We need to show: For any ε > 0, ∃ N such that:
| (n+1)/(3n-2) - 1/3 | < ε, ∀ n > N

Simplify:
= | [3(n+1) - (3n-2)]/[3(3n-2)] |
= | [3n+3 - 3n + 2]/[3(3n-2)] |
= | 5/[3(3n-2)] |

We need: 5/[3(3n-2)] < ε
⇒ 3(3n-2) > 5/ε
⇒ 3n-2 > 5/(3ε)
⇒ 3n > 5/(3ε) + 2
⇒ n > [5/(3ε) + 2]/3

Choose N > [5/(3ε) + 2]/3.

Then for all n > N, | (n+1)/(3n-2) - 1/3 | < ε.

Therefore, lim(n→∞) (n+1)/(3n-2) = 1/3 ✓
```

PROBLEM 5

Evaluate $\lim_{n \rightarrow \infty} (5n^2 + 3n + 3)/(2 - n + 7n^2)$

Solution:

Divide by n^2 :
 $= \lim_{n \rightarrow \infty} (5 + 3/n + 3/n^2)/(7 - 1/n + 2/n^2)$
 $= (5 + 0 + 0)/(7 - 0 + 0)$
 $= 5/7$

PROBLEM 6

Evaluate $\lim_{n \rightarrow \infty} 1/n^2 \times (n^2 - 1)/n^2$

Solution:

$= \lim_{n \rightarrow \infty} (n^2 - 1)/n^4$
 $= \lim_{n \rightarrow \infty} (1/n^2 - 1/n^4)$
 $= 0 - 0 = 0$

PROBLEM 7

Evaluate $\lim_{n \rightarrow \infty} \sqrt[3]{(3n^2 - 4n + 5)/(n + 1)}$

Solution:

Divide numerator and denominator by n :
 $= \lim_{n \rightarrow \infty} \sqrt[3]{(3 - 4/n + 5/n^2)/(1 + 1/n)}$
 $= \sqrt[3]{(3 - 0 + 0)/(1 + 0)}$
 $= \sqrt[3]{3}$

PROBLEM 8

Show that $\{(n+3)/(2n+1)\}$ is: (a) bounded, (b) monotonic decreasing. Find its limit.

Solution (a) Bounded:

$(n+3)/(2n+1) = 1/2 + (5/2)/(2n+1)$
For $n \geq 1$: $0 < 5/2/(2n+1) < 5/6$
So $1/2 < U_n < 1/2 + 5/6 = 4/3$
Therefore bounded.

(b) Monotonic decreasing:

$U_{n+1} - U_n = (n+4)/(2n+3) - (n+3)/(2n+1)$
 $= [(n+4)(2n+1) - (n+3)(2n+3)]/[(2n+3)(2n+1)]$
 $= [2n^2+9n+4 - (2n^2+9n+9)]/[(2n+3)(2n+1)]$
 $= -5/[(2n+3)(2n+1)] < 0$

So $U_{n+1} < U_n \rightarrow$ strictly decreasing.

(c) Limit:

$\lim_{n \rightarrow \infty} (n+3)/(2n+1) = \lim_{n \rightarrow \infty} (1 + 3/n)/(2 + 1/n)$
 $= (1+0)/(2+0) = 1/2$

PROBLEM 9

Find the limit points of the sequence defined by $a_n = 1 + (-1)^n + 1/n$

Solution:

When n is even: $(-1)^n = 1$
 $a_n = 1 + 1 + 1/n = 2 + 1/n \rightarrow 2$

When n is odd: $(-1)^n = -1$
 $a_n = 1 - 1 + 1/n = 1/n \rightarrow 0$

Therefore, the sequence has two subsequences converging to 2 and 0.

Limit points: $\{0, 2\}$

PROBLEM 10

Does the sequence $\{n!\}$ converge?

Solution:

$n! = 1 \times 2 \times 3 \times \dots \times n$
As n increases, $n!$ grows without bound.

$n=1$: 1
 $n=2$: 2
 $n=3$: 6
 $n=4$: 24
 $n=5$: 120
 $n=10$: 3,628,800

The sequence diverges to infinity.
So $\{n!\}$ does NOT converge.

Common Mistakes and How to Avoid Them

Mistake 1: Treating ∞ as a Number

Wrong	Correct
$\lim_{(n \rightarrow \infty)} (n+1)/(n) = \infty/\infty = 1$ ✗	$\lim_{(n \rightarrow \infty)} (n+1)/n = \lim_{(n \rightarrow \infty)} (1 + 1/n) = 1$ ✓

💡**TIP:** ∞ is NOT a number. You can't just substitute it!

Mistake 2: Applying Limit Rules Incorrectly

⚠️**CRITICAL WARNING:** $\lim_{(n \rightarrow \infty)} (U_n/V_n) = A/B$ even when $B = 0$ is WRONG. If $B = 0$ and $A \neq 0$, the limit does not exist. If $B = 0$ and $A = 0$, use other methods.

Mistake 3: Ignoring the Restriction for Infinite Sums

Wrong	Correct
$1 + 2 + 4 + 8 + \dots = 1/(1-2) = -1$ ✗	$1 + 2 + 4 + 8 + \dots$ diverges (no finite sum). $S_\infty = a/(1-r)$ ONLY works for $ r < 1$

Mistake 4: Confusing Sequence and Series

- **Sequence:** $\{U_n\} = 1, 1/2, 1/3, \dots$ (terms)
- **Series:** $\sum U_n = 1 + 1/2 + 1/3 + \dots$ (sum of terms)

💡**TIP:** A sequence can converge while its series diverges (e.g., $\sum 1/n$ diverges even though $1/n \rightarrow 0$).

Mistake 5: Forgetting to Check Monotonicity

When using the bounded monotonic theorem, you MUST check BOTH conditions!

Wrong	Correct
The sequence is bounded, so it converges. ✗	The sequence is bounded AND monotonic, so it converges. ✓

Quick Summary of Key Concepts

Sequences

- **Definition:** Ordered set of numbers with a rule
- **nth term:** U_n = formula
- **Convergent:** Has a finite limit
- **Divergent:** No finite limit

Arithmetic vs Geometric

Feature	AP	GP
Rule	$U_{n+1} = U_n + d$	$U_{n+1} = r \cdot U_n$
nth term	$a + (n-1)d$	$a \cdot r^{(n-1)}$
Sum	$n/2[2a+(n-1)d]$	$a(1-r^n)/(1-r)$
Infinity	Not applicable	$a/(1-r)$ if $ r <1$

Convergence Theorems

1. **Uniqueness of limit** — Only one limit
2. **Algebra of limits** — Combine limits with operations
3. **Bounded monotonic** — Bounded + monotonic $\rightarrow$ converges

Limit Evaluation

1. Divide by highest power for rational functions
2. Rationalize for expressions with roots
3. Use known limits ($1/n \rightarrow 0$, $(1+1/n)^n \rightarrow e$)
4. For exponential vs polynomial, exponential wins!